# Brendan King

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## Research Interests

I work toward open, adaptable **AI agents** that **interactively** help people solve complex problems, with a focus on **coding agents**, **dialogue interactions**, and **tool use**. My work spans benchmarks for dialogue-driven coding agents, methods for few-shot and unsupervised task-oriented dialogue, human-auditable confidence estimation for agents, and frameworks and corpora for multi-party dialogue structure.

## Education

**University of California, Santa Cruz** *Sept. 2021 - Present*
Ph.D. Candidate in Computer Science (Overall GPA: 4.00)
Advisor: Prof. Jeffrey Flanigan

**University of Washington, Seattle** *June 2017*
B.S. in Computer Science (Overall GPA: 3.63)

## Professional Experience

**Graduate Student Researcher & Instructor** September 2021 - Present
*UC Santa Cruz & NSF National AI Institute for Student-AI Teaming (NSF iSAT)*
*Advisor: Prof. Jeffrey Flanigan*

- Produced a new benchmark for evaluating **dialogue-driven coding agents**. [COLM WAB (2026)]
- Developed a state-of-the-art method for fully **unsupervised task-oriented dialogue** combining in-context learning, expectation-maximization, and noisy channel re-ranking. **[EMNLP (2024)]**
- Developed a state-of-the-art method for few-shot/zero-shot **dialogue state tracking** combining diverse retrieval-augmented in-context learning with **executable code actions**. **[ACL Findings (2023)]**
- Core contributor to the Dependency Dialogue Acts (DDA) **multi-party dialogue annotation scheme** **[IWSDS (2023)]**, and the first **DDA corpus and discourse parser** **[ACL Findings (2025)]**.
- **Graduate Student Instructor** for *Topics in Applied NLP* and TA for various NLP & ML courses.

**Research Scientist Intern** June 2026 - September 2026
*Megagon Labs, Inc.*
*Mentors: Farima Fatahi Bayat, Jean-Flavien Bussotti, Pouya Pezeshkpour, Estevam Hruschka*

- Developed an auditable inference-time confidence estimator for black-box **LLM agents**. [preprint (2026)]

**Software Development Engineer II** September 2017 - August 2021
*Apptio, Inc. - Platform Shared Services*

- Technical and project lead for a service coordinating end-to-end provisioning of all Apptio products from CRM data, reducing delivery time from days to minutes and cleaning up expired deployments.
- Technical lead for a service for managing versions and access for piral.io micro-frontends across products.
- Core maintainer of the authentication and authorization service for all Apptio products.

**Computational Research Intern** June 2015 - March 2017
*Institute for Systems Biology*
*Advisor: Prof. Nathan Price*

- Implemented the probabilistic annotation algorithm and its application to network gap-filling for constraint-based metabolic reconstructions as a python package, published in *Bioinformatics*. [paper]
- Designed and implemented algorithms for flux-preserving metabolic model translation from one organism to a genetic relative with improved preservation of evidence-based reactions, presented at *COBRA*.

## Selected Publications

**B. King**, F. Fatahi Bayat, J.-F. Bussotti, P. Pezeshkpour, E. Hruschka. Confidence Reasoning Graphs: Structured Confidence Estimation for LLM Agents. *arXiv preprint*, 2026. [preprint]

**B. King**, J. Flanigan. Dialogue SWE-Bench: A Benchmark for Dialogue-Driven Coding Agents. *COLM Workshop on AI Agent Behavior*, 2026. [preprint]

J. Z. Cai, **B. King**, P. Cameron, S. Brown, M. Eckert, D. Srinivas, G. Baker, V. K. Everson, M. Palmer, J. H. Martin, J. Flanigan. In Search of the Lost Arch in Dialogue: A Dependency Dialogue Acts Corpus for Multi-Party Dialogues. *Findings of the Association for Computational Linguistics (ACL)*, 2025. [paper] [corpus & codebook]

**B. King**, J. Flanigan. Unsupervised End-to-End Task-Oriented Dialogue with LLMs: The Power of the Noisy Channel. *Empirical Methods in Natural Language Processing (EMNLP)*, 2024. **Oral.** [paper] [code]

**B. King**, J. Flanigan. Diverse Retrieval-Augmented In-Context Learning for Dialogue State Tracking. *Findings of the Association for Computational Linguistics (ACL)*, 2023. [paper] [code]

J. Z. Cai, **B. King**, M. Perkoff, S. Dudy, J. Cao, M. Grace, N. Wojarnik, A. Ganesh, J. H. Martin, M. Palmer, M. Walker, J. Flanigan. Dependency Dialogue Acts – Annotation Scheme and Case Study. *International Workshop on Spoken Dialogue System Technology (IWSDS)*, 2023. **Oral.** [paper]

**B. King**, T. Farrah, M. A. Richards, M. B. Mundy, E. Simeonidis, N. D. Price. ProbAnnoWeb and ProbAnnoPy: Probabilistic Annotation and Gap-Filling of Metabolic Reconstructions. *Bioinformatics*, 2018. [paper]

## Teaching Experience

**Graduate Student Instructor**: Led *Topics in Applied NLP (Fall '26)* as a role-based reading seminar.

**Teaching Assistant:**
**Graduate:** Adv. Machine Learning for NLP (Win '23), Data Science and ML (Fall '23).
**Upper Div. Undergraduate:** Intro. Natural Language Processing (Spr '22, '23, '24).
**Lower Div. Undergraduate:** Programming Abstractions (Win '26), Intro. ML (Spr '26).

## Selected Awards and Honors

**Baskin Engineering Department Fellowship**, UC Santa Cruz 2026
**Outstanding Reviewer**, EMNLP - ACL Rolling Review 2024
**UC Regents Fellowship**, UC Santa Cruz 2021

## Service

**Reviewing:**
ICLR 2027
ACL Rolling Review *(top-100 Great Reviewer, 7 'great reviews')* 2024–2026
Workshops: COLM Agent Behavior, EACL SRW, ACL SRW, BayLearn 2025–2026

**Mentoring:**
UCSC CSE Directed Reading Program Mentor 2022

**Organizing:**
UCSC CSE PhD Slug Lunch Seminar 2025–2026

## Technical & Engineering Skills

Experienced software engineer and researcher, proficient with and without coding agents (Claude Code, Codex). Strengths and experience support ownership across the research-to-production pipeline:

**Research Engineering:** NLP and deep learning with Python, PyTorch, vLLM, OpenHands, AutoGen
**Service Design & Implementation:** owner of services in Python, Java, TypeScript/Node.js, React
**Deployment & Operations:** deploying production services with AWS, Docker, Kubernetes, CI/CD

## References

References are available upon request, please email me at bking2@ucsc.edu.